Hello, World!

1 Most common LaTeX commands

```
\def

Defines a new macro.
Syntax:
\def\<macro>{definition}
Example:
\def\square#1{#1^2} % Defines a macro square which squares its input.
When compared with \newcommand, note that \def is more primitive and doesn’t
\newcommand

A safer alternative to defining new macros than using plain TeX’s \def.
Syntax:
\newcommand{\<macro>}[num_args]{definition}
Example:
\newcommand{\myname}{John Doe}
\renewcommand
```

```
Used if you need to redefine an existing macro that was previously defined b
Syntax:
\renewcommand{\<macro>}[num_args]{definition}
\expandafter

Used for advanced TeX programming where controlling expansion timing matters
Example use case: Changing expansion order of macros before execution.
\csname ... \endcsname

Allows dynamic generation of control sequences.
Useful when you want programmatic access to macros whose names are not known
‘\if ... \fi Constructs These conditionals allow basic logic within LaTeX do

**\relax A no-op (does nothing). Often used as a placeholder or delimiter ma

Environment Handling (begin{} & end{}) Essential for creating structured blo
```

1. This will start math mode and line break examples:

Math mode:
The equation is $E = mc^2$

$$E = mc^2$$

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The equation is $E = mc^2$\$

$$E = mc^2$$

Math mode:
The equation
is $E = mc^2$

$$E = mc^2$$

We know from equation 1 that $E = mc^2$.
We know from equation 1 .... that $E = mc^2$.

$$E = mc^2 \tag{1}$$

We know from Equation 2:

$$E = mc^2 \tag{2}$$

2 This will start a new section with a number and title

This is the **bold** text.

3 This will start a new section with a number and title, but the title in the table of contents will be 'short title'

This is the *italic* text.

This will start a new section without a number and title

This is the <u>underlined</u> text.

4 Matrices and integrals expressions

First, import `amsmath package`.

4.1 Matrices 1

Here is a matrix:

$$A = \begin{pmatrix} a & b \\ c & d \\ 1 & 2 \end{pmatrix}$$

4.2 Integral 1

$$\int_0^\infty x^2 \, dx > \int_a^b x^2 * y \, dx \, dy$$

4.3 Multiple Integrals

This is a multiple integral that should be indented

$$\iiiint_V f(x, y, z, w) \, dx \, dy \, dz \, dw$$

This is a cyclic multiple integral (does not exist and does not make sense)

$$\oint_C f(x,y,z)\,dx\,dy\,dz$$

This is another multiple integral with fraction:

$$\iiiint_V \frac{f(x,y,z,w)}{2} x^2 y^2 + z^2 + w^2 \,dx\,dy\,dz\,dw$$

5 Derivatives

5.1 Partial Derivatives

to write partial derivatives:

$$\frac{\partial f}{\partial x_i}$$

5.2 using derivatives package

5.2.1 By using package derivatives

$$\frac{\partial^3 f}{\partial x_2\,\partial y^3\,\partial z} = \frac{\partial e^z \sin(x)}{\partial x} + \frac{\partial \frac{e^y}{\cos(y)}}{\partial y} + \frac{\partial e^z \cot(z)}{\partial z}$$

5.2.2 Evaluate at a specific point

Some example equations

$$\left(\frac{\partial U}{\partial x}\right)_p = \left.\frac{\partial U}{\partial x}\right|_p = \frac{\partial U}{\partial x}\Big|_{p,Y} \quad = \left.\frac{\partial U}{\partial x}\right|_{(p,Y)} \tag{3}$$

6 section Without Indentation

text2 should be indented

6.1 This is an unindented subsection

tex should be indented

6.2 Subsection With Indentation

This should be unindented

sdasdsda

7 Citation

This is a citation [1].

References

[1] Victor Hamann Pereira. *Desenvolvimento de método computacional para definição de rotas baseadas no uso do sistema de transporte coletivo*, Dec. 2017.